

MANOJ MUKHERJEE

AI Architect · Agentic AI & Multi-Agent Systems · Enterprise GenAI Platforms

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PROFESSIONAL SUMMARY

AI Architect and Engineering Leader with 10+ years delivering enterprise-scale AI systems, cloud-native platforms, and production GenAI applications across BFSI, Logistics, Cybersecurity, and Retail. Specializes in Agentic AI, multi-agent orchestration (LangGraph, Google ADK, MCP), and Enterprise RAG pipelines. Proven ability to drive 0 to 1 AI initiatives - from architecture and rapid prototyping through production deployment and scale, while partnering with C-suite stakeholders and mentoring high-performing engineering teams. Currently architecting AI platforms at Publicis Sapient that reduced manual workflows by 40%+ and accelerated delivery lifecycles by ~2x.

CORE SKILLS

Agentic AI	LangGraph, Google ADK, LangChain, MCP, Multi-Agent Orchestration, Agentic RAG, Tool Calling, Human-in-the-Loop
RAG & Context Engineering	Hybrid Search, pgvector, Reranking, Long-Term Memory, Context Engineering, Retrieval Evaluation
AI Reliability & Governance	LLM Evaluation, Responsible AI, Guardrails, AI Observability, Prompt/Context Testing
AI Platforms & Models	OpenAI, Anthropic Claude, AWS Bedrock, Vertex AI, Gemini, vLLM, Ollama
AWS AI Foundations	AWS AI/ML Services, Foundation Models, Generative AI, AWS Shared Responsibility Model
Backend & APIs	Python, FastAPI, Node.js, REST, GraphQL, WebSockets, Microservices
Data	PostgreSQL, MongoDB, pgvector, Pinecone, ChromaDB
Cloud & Platform Engineering	AWS, GCP, Kubernetes, Docker, OpenShift AI, NVIDIA Run:AI, CI/CD
Frontend Architecture	Next.js, React, TypeScript, Microfrontends, Module Federation
Architecture & Leadership	Enterprise AI Architecture, AI Strategy, System Design, Technical Governance, Stakeholder Management, Mentoring

AI ARCHITECTURE VALUE PROPOSITION

10+ yrs	3.5+ yrs	40%+	~2x
Enterprise software, cloud-native platforms, AI systems	Focused Enterprise GenAI, RAG, agents, LLM platforms	Manual workflow reduction via AI automation	Faster AI delivery lifecycle using reusable patterns

SELECTED AI PRODUCTS, CASE STUDIES & THOUGHT LEADERSHIP

Astra (অস্ট্রা) – AI Assistant Platform - <https://www.manojmukherjee.co.in/chat>

Designed and built a conversational AI assistant enabling users, engineers, students, architects, and business stakeholders to explore AI architectures, engineering practices, case studies, technical content, and consulting services through natural language interactions.

AI Gateway

Built a centralized AI Gateway architecture providing secure access to multiple LLM providers, AI services, and enterprise integrations through a unified interface.

GraphRAG & Enterprise Knowledge Systems

Designed retrieval architectures and GraphRAG systems enabling enterprise knowledge discovery, contextual reasoning, and AI-powered decision support.

Engineering & Architecture Lab

Created a public platform for sharing real-world system design patterns, AI architecture frameworks, engineering practices, and production implementation strategies.

AI Case Studies & Technical Research

Published practical case studies, engineering blogs, architecture deep dives, and implementation guides covering Agentic AI, Enterprise RAG, Multi-Agent Systems, AI Platform Engineering, and AI Transformation.

PROFESSIONAL EXPERIENCE

PUBLICIS SAPIENT | **AI Architect / Technical Lead** *Mar 2023 – Present · Bengaluru*

- Architected enterprise Agentic RAG platforms using LangGraph, MCP, and multi-agent orchestration, serving complex BFSI workflows for Fortune 500 clients.
- Designed multi-agent systems with dynamic tool calling, long-term memory, and contextual retrieval pipelines using pgvector + MongoDB hybrid search, improving retrieval relevance by 35%+.
- Built production AI microservices (FastAPI, containerized Python) and full-stack AI applications (Next.js, TypeScript, AI SDKs) deployed on Kubernetes, OpenShift AI, and NVIDIA Run:AI.
- Led AI architecture reviews, discovery workshops, and strategy sessions with enterprise stakeholders and senior leadership (CTO/VP level).
- Conducted rapid prototyping for RFPs across BFSI, Retail, and Logistics; accelerated POC-to-production lifecycle by nearly 2x.
- Mentored engineers in full-cycle AI engineering (design → deploy → scale), delivered AI workshops and L&D programs across the org.
- Reduced manual workflow effort by 40%+ via AI-driven automation; received FS WEST SUPERNOVA Award and Spot Award for AI Innovation.

KOTAK MAHINDRA BANK | **Chief Manager (SDE III)** *Sep 2022 – Feb 2023 · Bengaluru*

- Contributed to architecture initiatives for Kotak811 digital banking platform, focusing on fraud prevention, compliance, security, and UPI 2.0 modernization.
- Collaborated with CTO and business stakeholders on product and architecture decisions; guided external/vendor engineering teams on standards and best practices.

A.P. MØLLER – MAERSK | **Software Engineer III** *Aug 2021 – Aug 2022 · Bengaluru*

- Established microfrontend architecture using Module Federation (React + Next.js), enabling independently deployable frontend modules at enterprise scale.
- Implemented SSR for microfrontends, improving performance by 25–30% and reducing initial load times; shaped the Anchor Design System across Maersk's React ecosystem.
- Mentored engineers and provided architectural direction across teams; received Maersk Star Award.

KRISTA SOFTWARE | **Senior Software Engineer / Founding Engineer** *Jun 2020 – Jul 2021 · Bengaluru*

- Founding engineering member for an AI-driven enterprise automation platform; contributed to architecture, scalability strategy, and cross-platform product delivery.
- Led functional development, code reviews, and mentored engineers; received Krista Excellence Award for outstanding impact on engineering team.

OPTIV INC | **Software Engineer** *Jun 2019 – Jun 2020 · Bengaluru*

- Built cybersecurity platforms, analytics dashboards, and internal APIs using React.js, GraphQL, Node.js, and MongoDB; contributed to CI/CD pipeline and production support.

HEWLETT PACKARD ENTERPRISE | **Software Engineer I** *Dec 2018 – Jun 2019 · Bengaluru*

- Built React.js / Redux interfaces for hybrid cloud products (OpenSphere, GreenLake); contributed to open-source Grommet design system and gained cloud-native infrastructure foundations.

WILLIAM O'NEIL INDIA | **Software Engineer** *Jul 2016 – Dec 2018 · Bengaluru*

- Built Panaray, a flagship financial analytics and real-time market intelligence platform, from scratch—including frontend architecture, state management (Flux → Redux/Saga), and BFF services.
- Mentored junior engineers; multiple mentees progressed to senior/tech-lead roles. Received Excellence Award for Rapid Delivery.

KEY AI PROJECTS & PRODUCTS

Enterprise Agentic RAG Platform (BFSI)

- Designed enterprise-grade multi-agent architecture for BFSI workflow automation using LangGraph and MCP integrations.
- Implemented hybrid retrieval with pgvector + MongoDB, long-term memory, and contextual reasoning pipelines; significantly improved retrieval relevance and generative response quality.
- Deployed scalable AI infrastructure on Kubernetes and OpenShift AI; established reusable AI platform patterns across projects.

AI Gateway — Unified Multi-LLM Enterprise Gateway

- Architected a unified gateway abstracting OpenAI, Claude, Gemini, and self-hosted LLMs (vLLM, Ollama) behind a single FastAPI interface with routing, fallback, and rate-limiting.
- Enabled enterprise teams to swap LLM providers without code changes; accelerated GenAI experimentation cycles.

Astra — Personal AI Architect Assistant

- Built a production AI assistant powering engineering consulting, architecture case studies, and discovery (manojmukherjee.co.in).
- Leverages Agentic RAG, MCP tools, and contextual memory; serves as a live demonstration of production-grade AI architecture patterns.

Vision AI & OCR Intelligence System

- Built AI systems integrating OCR, Vision LLMs, and generative models for intelligent document extraction and contextual reasoning at scale.

PUBLICATIONS & THOUGHT LEADERSHIP

Published on Medium (<https://www.medium.com/@manojmukherjee777>) — selected technical writing:

- Agent-to-Agent Communication Systems and orchestration patterns
- Advanced Retrieval-Augmented Generation (RAG): hybrid search, memory, and contextual retrieval
- Building Real-Time AI Agents with LangChain, LangGraph, and Open Source LLMs
- AI + OCR + Vision Systems: intelligent document extraction pipelines
- AI-Powered Stock Analysis using LangGraph & DeepSeek

EDUCATION

Master of Computer Applications (MCA) 2013 – 2016 · 80%

Visvesvaraya Technological University, India

Bachelor of Computer Applications – Honours (BCA) 2010 – 2013 · 76% · University 3rd Rank

University of Burdwan, India

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **AWS Certified AI Practitioner** <https://www.credly.com/badges/7ea65b79-ee4d-449f-8d98-f52fd0734b88>
- **AWS Partner: Generative AI Essentials - Training Badge**
<https://www.credly.com/badges/d73299c5-9799-40e3-aa36-3154e0e85933>
- **Google AI Leadership** | Udemy coursework completed; Google certification exam planned
- **Applied Quantum AI PhD** | Planned pursuit from Jan 2027 (prospective; not yet enrolled)

AWARDS & RECOGNITION

FS WEST SUPERNOVA Award · Publicis Sapient · Industry recognition for AI innovation and thought leadership

Spot Award for AI Innovation & Mentoring · Publicis Sapient

Maersk Star Award · A.P. Møller – Maersk · Engineering excellence and architectural impact

Krista Excellence Award · Krista Software · Outstanding engineering impact as Founding Engineer

University 3rd Rank Holder · BCA Honours, University of Burdwan

ADDITIONAL

Languages: English (Professional) · Hindi (Professional) · Bengali (Native)

Open to: Worldwide engagements · Relocation · Visa sponsorship

Architecture Lab & Case Studies: manojmukherjee.co.in/case-studies